Storybook Generator

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1. Project Overview

This project focuses on designing and developing a Kids' Storybook Generator that collects its required information through a chatbot. The goal of this project is to demonstrate to the user how visually appealing art can be created easily and fully automated through AI.

2. Problem Statement

2.1. Role

Storybooks and stories play a foundational role in our lives. They instill the foundational morality through examples in the form of stories. The morality according to which we decide right from wrong and truth from falsehood.

2.2. Need

There is a need for a convenient way to have access to stories on a desired topic with beautiful art and visual appeal for children's interest.

2.1. Literature Review

Chatbots like ChatGPT can write a story and generate images, but the process is not fully automated and user-friendly at all. Storyline has to be described without the chatbot guiding you unless you gave it a prompt to do so. Images can be generated for your story, but they have to be described in detail according to your current page of the story. Not to mention image generation models on chatbots like ChatGPT can generate a very limited number of images or are completely paid. And on top of all that, you have to compile the PDF yourself.

3. Proposed Solution

To address this problem, we developed an AI-based storybook generator with a chatbot:

- Chatbot converses with the user for the required information
- Sends that information to a book generation system
- System writes a book with beautiful images (Text-on-Image)
- System returns the generated book to the user in the chat

4. Tools and Technologies Used

- **AI Tools:** Groq
- **LLM:** gpt-oss-120b
- **Cloud Hosting:** Modal
- **Image Generation Model:** Z-Image-Turbo(fast)
- **Programming Language:** JavaScript
- **Automation Tool :** n8n
- **Chatbot Frontend:** AI Agent Node n8n
- **Storage:** Google Drive

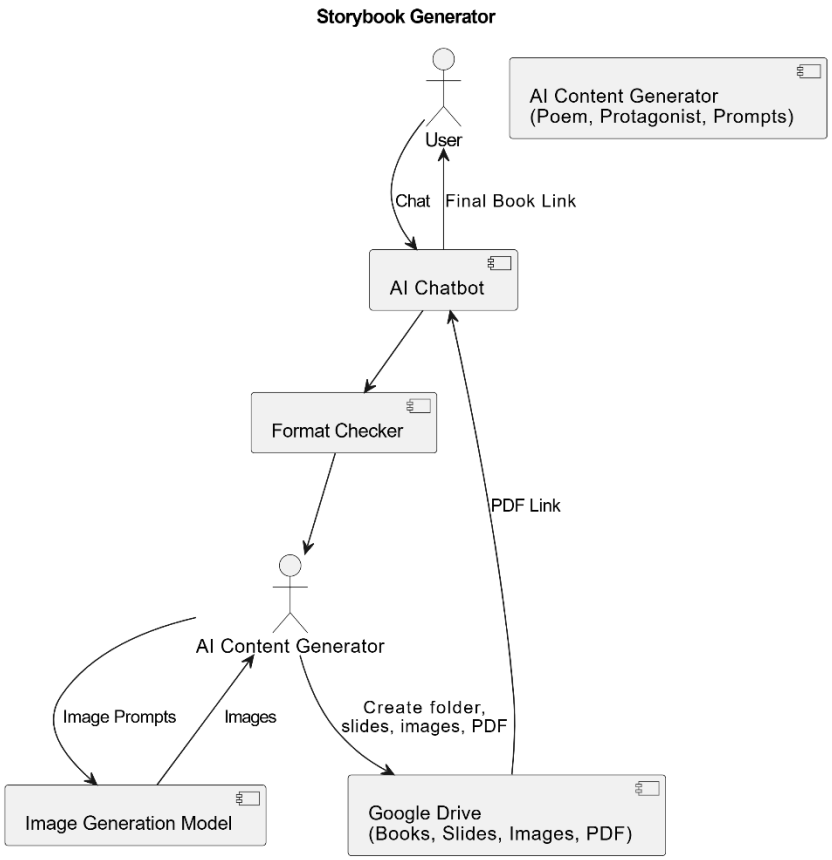
5. System Workflow

5.1. Workflow

1. User converses with the AI chatbot
2. The chatbot interprets user responses and answers accordingly
3. Required information is passed by the chatbot to a format checker
4. Format checker then passes it to the relevant nodes
5. Relevant node creates the folder for the book in the 'books' folder on Google Drive
6. A slide is created in the folder for the book on Google Drive from a template in Google Drive's 'books' folder called 'Book_Template.pptx'
7. The protagonist and poem are created by AI simultaneously with the folder
8. Prompt for book cover (image) is created according to the poem and the protagonist
9. Prompt is sent to the Image generation model
10. Image is generated by the model in base64
11. Image is converted into a binary file (PNG)
12. Image is uploaded to Google Drive
13. Image permissions on Google Drive are changed to make it viewable for anyone with the link
14. Image of book cover is converted back to base64 for reference to further nodes (now redundant)
15. In the same fashion, all the image generation prompts for all the pages (images) are generated by AI, given to the image generation model, received in base64, converted to binary, and uploaded to Google Drive.

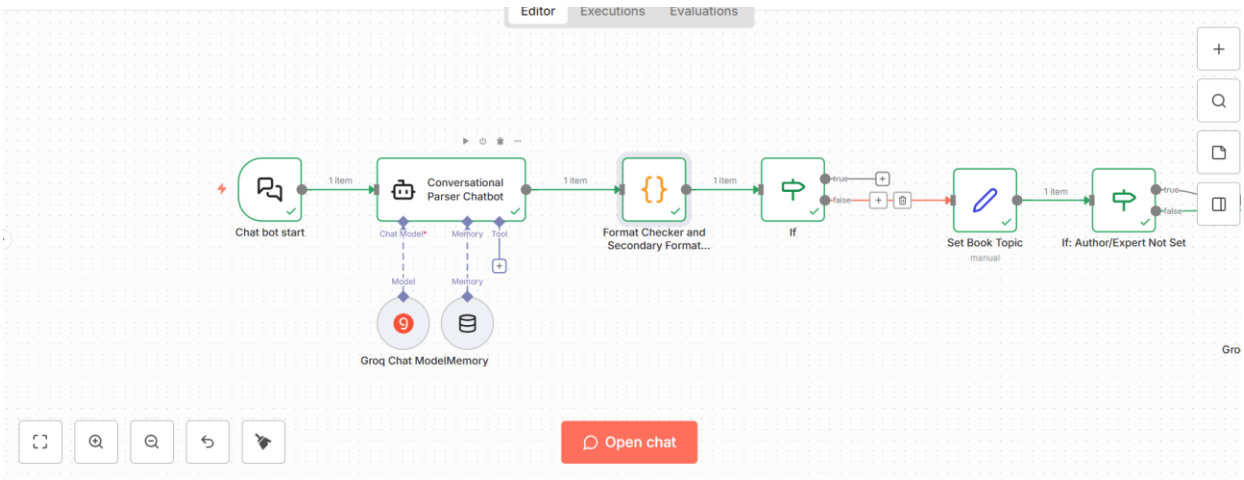
16. Pages (Images) are sorted, permissions on Google Drive are changed to make it viewable for anyone with the link, and added to the slide created from the template in Google Drive
17. Slide is converted to PDF
18. PDF is uploaded to the book folder
19. PDF permissions on Google Drive are changed to make is viewable for anyone with the link
20. Link of the final generated book in PDF format, stored in the book folder in the 'books' folder, is given to the user in the chat

5.2. System Architecture

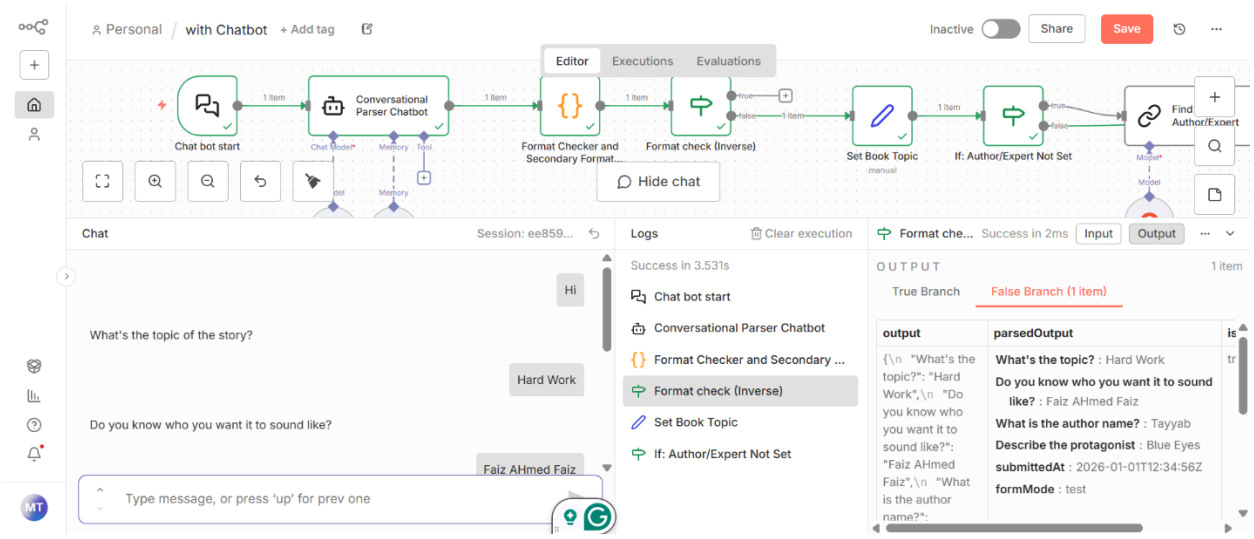


5.3. Testing

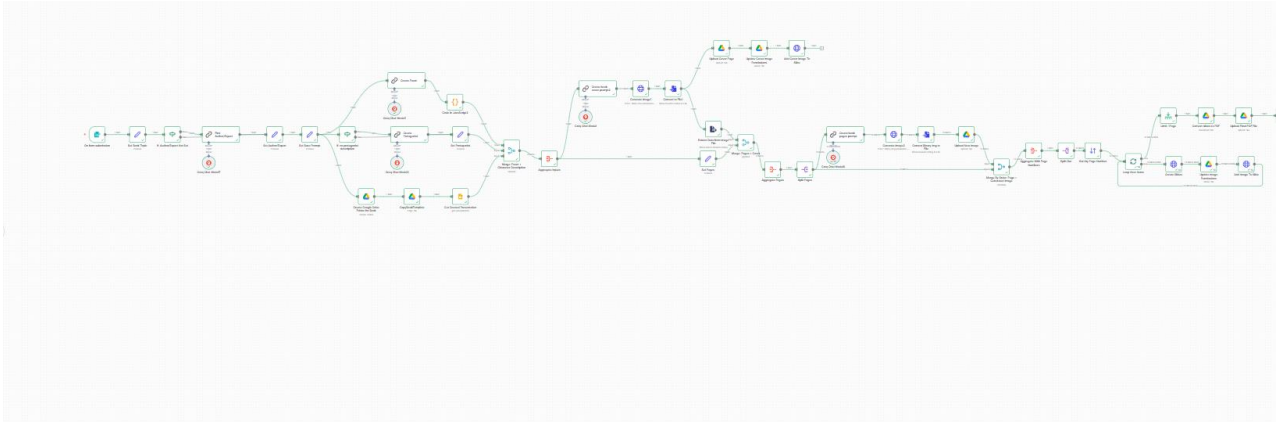
- Chatbot:



- Chatbot responses:

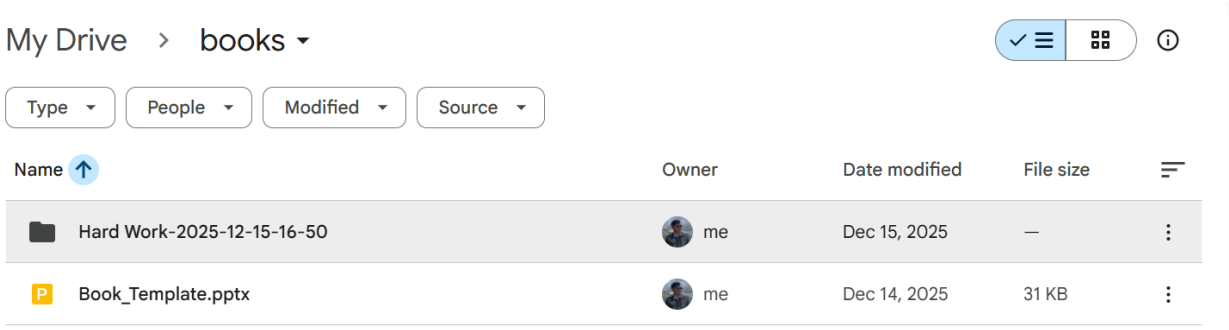


- Generation system:



Note: Rest of the workflow executed and tested before the integration of chatbot (all nodes are green).

- Google Drive ‘books’ folder



- Book folder named after topic name on Google Drive

6. Key Features

- Fast response time
- User-friendly interface
- AI-powered automation
- Intelligent responses
- Human-like understanding and memory

- Accurate image and story generation
- Auto compilation
- High-quality production

7. Individual Contributions

- **Abdul Mateen:** Idea generation and project planning
- **Muhammad Tayyab:** Workflow Automation, AI integration, and deployment
- **Muhammad Adan Ch:** Documentation and presentation
- **Areel Ashfaq:** Testing, debugging.

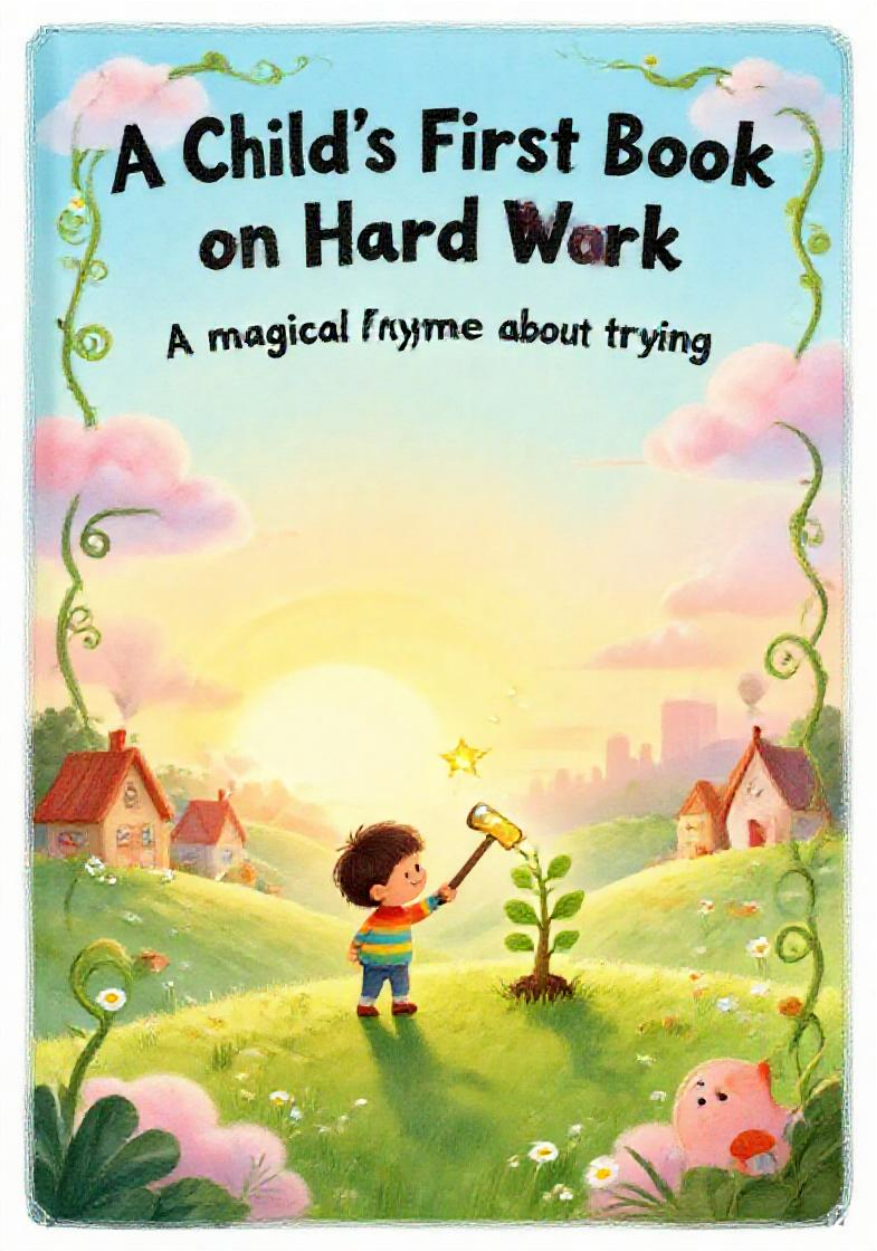
8. Challenges Faced

- Handling user queries
- Handling AI outputs
- Integrating AI tools
- Prompt engineering system prompts of AI agents
- Free image generation
- Selection of the right free image generation model
- Free hosting of image generation models
- Accurate image generation
- Ensuring good Text-on-Image Quality as pages are just images with text on them
- Ensuring accurate responses
- Storage management
- File handling
- File management (upload/download) on Google Drive
- File consistency
- Resource availability
- Format conversion both text and file
- Error Handling

9. Learning Outcomes

9.1 Expected and Achieved Outcomes

The expected and achieved outcomes are the same, that is, a book PDF in Google Drive with its link in the chat. Link given below:
https://drive.google.com/file/d/154WMaInlxfr2GuRPhWQUABL1P7rW3BiV/view?usp=drive_open



9.2 Learning Outcomes

Through this project, we gained:

- Practical understanding of AI-based systems
- Experience with automation tools especially n8n
- Knowledge of chatbot development
- Knowledge of LLMs
- Knowledge of Image Generation Models
- Knowledge of self/cloud hosting AI models both image generation and LLMs
- Use of APIs
- Team collaboration and project management skills

10. Conclusion

This project successfully demonstrates the use of AI tools to solve real-world problems. The AI chatbot enhances the user-friendliness by automating query handling. The Storybook generation system is an efficient way to get the required book in its final form by automating the generation and compilation process.